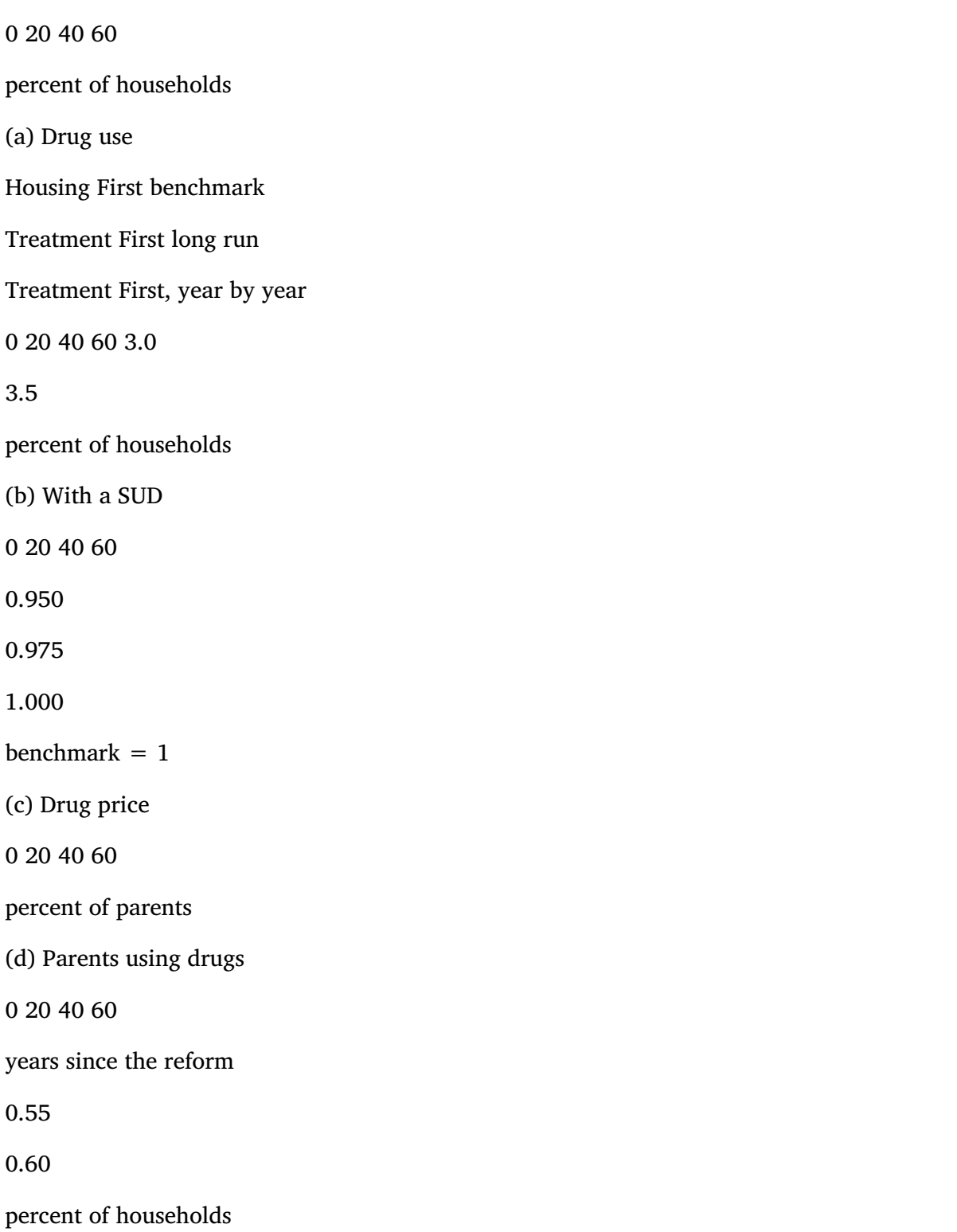
The main transition lets a household with a SUD keep its voucher by entering treatment and not using. This section asks what the first periods of the reform look like when households with a SUD cannot stop using,



(e) Homeless

0 20 40 60

years since the reform

0.95

1.00

percent of children

(f) Children in a shelter

0 20 40 60

year of birth, after the reform

percent of lifetime consumption

(g) Welfare by year of birth, CEV

0 20 40 60

years since the reform

percent of households

(h) Worse off than under Housing First

all households

households with a SUD

Figure 2: The transition from Housing First to Treatment First, quantitative model

Notes: panels (a) to (f) and (h) are by year since the reform, which lands at the start of year 1, with the benchmark as year 0; the dashed orange line is the Housing First benchmark, the dotted blue line the Treatment First long run and the solid blue line the path. In panel (h) the dashed line is the share of households with a SUD who are worse off. Children are those of households aged 26 to 50. Panel (g) is the CEV of each cohort born on the path, against the benchmark, by its year of birth. Panel (h) is the share of households worse off in each year of the path than in the same state under Housing First, weighted by the benchmark's distribution.

so that only after recovery can they stay sober and pass the screen. I consider three alternatives, each with the screen catching every user or catching 20 percent of users, and with the drug price and πV clearing every period. In the first, the screen applies from day one of the reform, and the disorder fails it whether or not the household is in treatment. In the second, the reform is announced, but Housing First continues for five periods and the same screen starts in period six. In the third, the same screen applies from the reform, but for its first five periods a household in treatment keeps its voucher even if it uses drugs.

Table 7 shows how homelessness, the share of children in a shelter and the welfare of children born at the reform compare with the Housing First benchmark in each, beside the main transition (first row).

Table 7: The first years of the alternative transitions

Homelessness Children in a shelter Newborn’s

pdet Largest rise Periods above Largest rise Periods above gain

Main transition 20 none none none none 1.45

No exemption

100 13.6 (2) 1–15 10.8 (2) 2–11 6.74 20 – – – – –

Announced five periods ahead

100 7.3 (9) 7–16 3.6 (9) 7–12 4.94 20 3.1 (12) 7–19 0.6 (11) 9–13 2.57

Sobriety while in treatment exempt for the first five periods

100 4.2 (9) 7–13 0.2 (9) 8–9 6.88 20 1.5 (11) 9–15 none none 3.14

Notes: Each row is a transition from the Housing First benchmark to Treatment First with the drug price and πV clearing every period, against Housing First’s steady state. Main transition: Section 6.4, where a household with a SUD keeps its voucher by entering treatment and not using. In the other rows it cannot stop using, and the disorder fails the screen even in treatment: from the reform (no exemption), from period six with Housing First until then (announced five periods ahead), or from period six with treatment alone passing it until then (sobriety while in treatment exempt for the first five periods). pdet is the detection rate, in percent. Largest rise: the largest percent increase over the first fifteen periods, with its period in brackets. Periods above: the periods in which it is above Housing First. Children in a shelter is the homelessness of households aged 26 to 50. Newborn’s gain: the CEV of children born at the reform, percent of lifetime consumption. When every user is caught, no drug price clears the long-run drug market, because drug use jumps as the price moves through the level that would clear it; those transitions end at the long-run solution that comes closest.

In the first alternative, how much the screen catches decides what happens along the way. Catching every user lowers homelessness by 30 percent and the share of children in a shelter by 31 percent in the long run, but it puts every voucher holder with a SUD out at once: homelessness rises by up to 14 percent and the share of children in a shelter by up to 11 percent, and they stay above their Housing First levels for 15 and 11 years, respectively, because πV rises only as the savings accrue. Catching only one in five users lowers both in the long run too, homelessness by 11 percent and the share of children in a shelter by 11 percent, but it also raises them in the first years, though by less, because it puts fewer households out at once: before its savings are recycled and the drug price falls, homelessness rises by up to 6 percent and the share of children in a shelter by up to 4 percent. Even so, children born at the reform gain 6.74 percent of lifetime consumption when every user is caught and – percent when 20 percent are, against 1.45 percent in the

main transition. They gain more because the screen deters more: it catches every user, and a household with a SUD cannot stop using, so the disorder costs it its voucher until it recovers, while in the main transition the household keeps the voucher by entering treatment and not using. By period ten, 2.80 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder, and πV is higher throughout their childhood.

Announcing the screen five periods before it starts lowers the rise but does not eliminate it. While Housing First continues, households cut their drug use at once in anticipation of a screen that will catch every user, by 31 percent in the first period, and homelessness and the share of children in a shelter fall below their Housing First levels. Once it starts revoking vouchers, they rise by up to 7 and 4 percent,

against 14 and 11 percent without the announcement. When it catches one in five users, they rise by up to 3 and 0.6 percent. Children born at the reform gain 4.94 percent of lifetime consumption when every user is caught and 2.57 percent when one in five users are. The first is less than the 6.74 percent without the announcement, because πV does not rise until the screen starts and drug use falls by less in the first period than the 40 percent when the screen starts at once. The second is also more than the 1.45 percent in the main transition, although the screen catches one in five users in both. Here a household with a SUD cannot stop using, so once the screen starts the disorder puts its voucher at risk until it recovers, while in the main transition the household keeps the voucher by entering treatment and not using. Starting to use therefore risks more, and fewer households use drugs: by period ten, 3.27 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder. Exempting sobriety while in treatment for the first five periods lets a household with a SUD keep its voucher in those periods by entering treatment, even if it uses drugs, and more of them do: 36 percent of those with a SUD are in treatment, against 25 percent under Housing First. When the screen catches one in five users, homelessness stays below its Housing First level for the first eight periods but rises above it by at most 1.5 percent after the exemption ends, and the share of children in a shelter never rises above its Housing First level. When every user is caught, homelessness stays below its Housing First level for the first six periods but rises above it by at most 4.2 percent after the exemption ends, and the share of children in a shelter by at most 0.2 percent. Children born at the reform gain 3.14 percent of lifetime consumption when one in five users are caught, more than the 1.45 percent in the main transition, because once the exemption ends a household with a SUD, which cannot stop using, risks its voucher until it recovers, so fewer households use drugs: by period ten, 3.22 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder, and πV is higher throughout their childhood. When every user is caught, they gain 6.88 percent.

7 Discussion and implications

To close, I return to the question I started with: should housing assistance require sobriety? The answer turns on the children, and not on the moral hazard that dominates the policy debate: without the externalities on children, Treatment First cuts drug use by 28 percent, but a newborn is worse off, because the screen takes an option from households that weigh the risks correctly and gives back less through the budget than it takes.

Drug use costs children consumption and housing at home and productivity in adult life, and a parent counts only the share ωc of either, so the screen's deterrence is worth more to the child than to the parent. With the drug price and πV held, the parents lose 1.19 percent of their own lifetime consumption, about what they lose without the externalities, while a newborn gains 2.82 percent, so the reform raises welfare on net once the children are counted. With the budget rebalanced and the drug price clearing, as they would be after a real reform, Treatment First cuts drug use by 21 percent, lowers homelessness by 11 percent, and raises a newborn's welfare by 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. Once general-equilibrium effects are accounted for, the parents gain too, because the higher πV and the cheaper drug outweigh the option the screen removes. Along the transition, neither homelessness nor the share of children in a shelter rises, and children born at the reform gain 1.45 percent.

Three lessons follow. First, the arguments for Housing First and Treatment First made by different advocacy groups leave out the outcome that matters most: what happens to the children of the households a sobriety requirement deters from using, in what they consume while living with their parents, in whether they become homeless and in what they earn as adults.

Second, the model agrees with the micro evidence behind Housing First on the households it follows: housing, and a sobriety condition on it, raise the recovery of households with a SUD by about as much as in the trial that measures both, and losing a voucher raises the mortality of households with SMI by about as much as Housing First lowers it among homeless veterans with mental illness. As that evidence implies, those with a SUD lose from Treatment First: with the drug price and πV held, as in a trial, which is too small to move either, – percent of those who have one when the reform lands are worse off, and even once both clear, 67 percent are. The macro effects the trials do not capture, however, overturn the conclusions.